Integrated AI Query System

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October 19, 2025

Project Structure and Execution Summary

This report documents the implementation of an Integrated AI Query System designed to unify and query multiple data sources including project developers, investors, emails, and meeting transcripts. The system provides intelligent search capabilities across all datasets with robust data cleaning, entity matching, and semantic search functionality.

1 Project Directory Structure

```
Calculus_Carbon_Assignment/
          data/                           # Data directory
                Project_developer.csv       # Raw project developer data
                Investors.csv              # Raw investor data
                Outlook_emails.csv         # Raw email data
                Meeting_transcripts.csv    # Raw meeting transcripts
                clean_developers.csv       # Cleaned developer data
                clean_investors.csv        # Cleaned investor data
                clean_emails.csv           # Cleaned email data
                clean_meetings.csv         # Cleaned meeting data
                unified_matches.csv        # Project-investor matches
                search_index.pkl           # Semantic search index
          src/                            # Source code directory
                __init__.py                # Package initialization
                data_clean.py              # Data cleaning and integration
                build_index.py             # Semantic search index builder
                query_layer.py             # Query processing system
                app_streamlit.py           # Web interface
          main.py                        # Main setup script
          requirements.txt               # Python dependencies
```

1.1 Run Instructions

The project includes an automated setup script that handles all necessary steps:

```
# Run the main setup script - it will handle everything automatically
# Ensure The Virtual Environment is Running
python main.py
```

What the automated setup does:

1. Checks and installs required dependencies from `requirements.txt`
2. Executes data cleaning and integration (`data_clean.py`)
3. Builds the semantic search index (`build_index.py`)

4. Launches the Streamlit web application
5. Provides status updates for each step

2 Data Integration & Cleaning

2.1 Data Loading and Initial Processing

The system begins by loading four primary data sources from CSV files:

```python
def load_data(self):
        Load all data sources from data/ directory
    self.developers_df = pd.read_csv('data/Project_developer.csv', skiprows=1)
    self.investors_df = pd.read_csv('data/Investors.csv', skiprows=1)
    self.emails_df = pd.read_csv('data/Outlook_emails.csv', skiprows=1)
    self.meetings_df = pd.read_csv('data/Meeting_transcripts.csv', skiprows=1)
```

Listing 1: Data Loading Implementation

Key Implementation Details:

- `skiprows=1` handles header rows in raw data
- Automatic removal of unnamed columns and empty rows
- Text normalization using custom `clean_text()` function

2.2 Project Developers Data Cleaning

```python
# Standardize inconsistent project type naming
project_type_mapping = {
    'ARR': 'Afforestation/Reforestation',
    'AWD (rice)': 'Alternative Wetting&Drying',
    'Methane Leak Repair': 'Methane Reduction',
    'Biochar pilot': 'Biochar',
    'Mangrove Restoration': 'Blue Carbon'
}
self.developers_df['ProjectType'] =
    self.developers_df['ProjectType'].replace(project_type_mapping)

# Create region mapping from countries
region_mapping = {
    'Brazil': 'Latin America', 'Peru': 'Latin America',
    'Philippines': 'Asia', 'Vietnam': 'Asia',
    'Senegal': 'Africa'
}
self.developers_df['Region'] =
    self.developers_df['Country'].map(region_mapping)
```

Listing 2: Project Type Standardization

Handling Missing Data:

- Numeric fields converted with error handling: `pd.to_numeric(errors='coerce')`
- Status fields filled with 'Unknown' for missing values
- Region mapping ensures consistent geographical categorization

2.3 Investors Data Cleaning

```
# Consolidate sector focus categories for better matching
sector_mapping = {
    'NbS, ARR, REDD+': 'ARR/REDD+',
    'ARR, AWD': 'ARR/Agriculture',
    'Agriculture, Restoration, Agroforestry': 'Agriculture/Agroforestry'
}
self.investors_df['SectorFocus'] =
    self.investors_df['SectorFocus'].map(sector_mapping)

# Validate and convert ticket size data
self.investors_df['TicketSizeMin'] = pd.to_numeric(
    self.investors_df['TicketSizeMin'], errors='coerce'
)
self.investors_df['TicketSizeMax'] = pd.to_numeric(
    self.investors_df['TicketSizeMax'], errors='coerce'
)
```

Listing 3: Investor Data Standardization

Data Quality Improvements:

- Region focus standardized with 'Global' as default for missing values
- Ticket sizes validated and converted to numeric format
- Sector categories consolidated for improved matching accuracy

2.4 Emails Data Processing

```
# Extract project references from email bodies
project_keywords = [f'P{i:03d}' for i in range(1, 16)]

def extract_project_references(text, project_keywords):
        Extract project references from email text
    if pd.isna(text):
        return []
    text = str(text)
    found_projects = [proj for proj in project_keywords if proj in text]
    return found_projects
self.emails_df['ReferencedProjects'] = self.emails_df['Body'].apply(
    lambda x: extract_project_references(x, project_keywords))
```

Listing 4: Project Reference Extraction from Emails

Entity Extraction Features:

- Project IDs (P001-P015) automatically detected in email content
- Date fields converted to datetime format for temporal analysis
- Text cleaning removes extra whitespace and normalizes format

2.5 Meeting Transcripts Processing

```
def extract_entities(text):
        Extract projects, investors, and actions from meeting text
    entities = {'projects': [], 'investors': [], 'actions': []}

    # Map developer names to project IDs
```

```
    project_mapping = {
        'VerdeNova': 'P001', 'EquiForests': 'P002', 'BlueCanyon': 'P003'
    }

    for dev, proj in project_mapping.items():
        if dev in text:
            entities['projects'].append(proj)

    # Extract action items using keyword matching
    action_keywords = ['schedule', 'visit', 'share', 'prepare', 'review']
    for action in action_keywords:
        if action in text.lower():
            entities['actions'].append(action)

    return entities

self.meetings_df['ExtractedEntities'] =
    self.meetings_df['TranscriptText'].apply(extract_entities)
```

Listing 5: Entity Extraction from Meeting Transcripts

Advanced NLP Features:

- Named Entity Recognition for projects and investors
- Action item extraction from natural language using keyword matching
- JSON serialization for structured storage and retrieval

2.6 Unified Dataset Creation

```
def calculate_match_score(project, investor):
        Calculate compatibility score between project and investor
    score = 0
    match_reasons = []

    # Region matching (30% weight)
    project_region = project.get('Region', '')
    investor_region = investor.get('RegionFocus', '')

    if investor_region == 'Global' or project_region in str(investor_region):
        score += 0.3
        match_reasons.append(f Region alignment: {project_region} )

    # Sector matching (40% weight)
    project_type = project.get('ProjectType', '')
    investor_sector = investor.get('SectorFocus', '')

    if any(keyword in investor_sector for keyword in project_type.split()):
        score += 0.4
        match_reasons.append(f Sector alignment: {project_type} )
    # Ticket size matching (30% weight)
    project_scale = project.get('Hectares', 0) or 0
    project_scale_value = project_scale * 1000
    ticket_min = investor.get('TicketSizeMin', 0) or 0
    ticket_max = investor.get('TicketSizeMax', float('inf')) or float('inf')
    if ticket_min <= project_scale_value <= ticket_max:
        score += 0.3
        match_reasons.append( Ticket size appropriate )
    return round(score, 2), match_reasons
```

Listing 6: Project-Investor Matching Algorithm

Matching Algorithm Features:

- Multi-factor scoring with weighted components
- Threshold-based filtering (score ¿ 0.3) for relevant matches
- Match reasons tracking for explainable AI and transparency

3 Semantic Search Implementation

3.1 Vector Index Building

```
class VectorIndexBuilder:
    def __init__(self):
        # Use sentence transformer for semantic embeddings
        self.model = SentenceTransformer('all-MiniLM-L6-v2')

    def build_index(self, documents):
            Build semantic search index from all documents
        texts = [doc['text'] for doc in documents]
        # Generate embeddings with normalization
        self.embeddings = self.model.encode(texts, normalize_embeddings=True)

        # Save index for fast loading
        index_data = {
            'embeddings': self.embeddings,
            'documents': documents
        }
        with open('data/search_index.pkl', 'wb') as f:
            pickle.dump(index_data, f)
```

Listing 7: Semantic Search Index Construction

Technical Implementation Details:

- Uses `all-MiniLM-L6-v2` sentence transformer for high-quality embeddings
- Embeddings normalized for accurate cosine similarity calculations
- Index serialized using pickle for fast application startup

3.2 Document Preparation

```
def prepare_documents(self):
        Prepare documents from all data sources for indexing
    documents = []

    # Developer documents with structured metadata
    for _, row in self.developers.iterrows():
        doc_text = f
        Project Developer: {row['DeveloperName']}
        Project ID: {row['ProjectID']}
        Project Type: {row['ProjectType']}
        Country: {row['Country']}
        Region: {row.get('Region', 'N/A')}
        Status: {row['Status']}

        documents.append({
            'text': doc_text,
            'source': 'developers',
            'id': row['ProjectID'],
```

```
            'type': 'project',
            'metadata': {
                'name': row['DeveloperName'],
                'project_type': row['ProjectType'],
                'region': row.get('Region', '')
            }
        })
    # Similar processing for investors, emails, and meetings
    return documents
```

Listing 8: Multi-source Document Preparation

Multi-source Integration Features:

- Structured metadata for advanced filtering and provenance tracking
- Unified text representation enabling cross-source semantic search
- Source and type tracking for data lineage and result explanation

4 Query Layer Architecture

4.1 Core Query System

```
class QuerySystem:
    def __init__(self):
        self.model = SentenceTransformer('all-MiniLM-L6-v2')
        self.load_data()  # Load index and cleaned data

    def semantic_search(self, query: str, k: int = 5, source_filter: str =
        None):
            Perform semantic search across all documents
        query_embedding = self.model.encode([query], normalize_embeddings=True)
        similarities = cosine_similarity(query_embedding, self.embeddings)[0]

        # Get top results with filtering
        top_indices = np.argsort(similarities)[::-1][:k*2]
        results = []
        for idx in top_indices:
            result = self.documents[idx].copy()
            result['score'] = float(similarities[idx])

            if source_filter and result['source'] != source_filter:
                continue
            results.append(result)

        return results[:k]
```

Listing 9: Semantic Search Implementation

Search Capabilities:

- Cosine similarity for accurate semantic matching
- Source-based filtering for targeted searches
- Configurable result limits with oversampling for filtering

4.2 Intelligent Question Parsing

```python
def answer_question(self, question: str):
        Parse natural language questions and route to appropriate handlers
    question_lower = question.lower().strip()

    # Pattern 1: ARR projects in specific regions
    if any(pattern in question_lower for pattern in [
        'developers have arr projects in latin america',
        'arr projects latin america',
        'latin america arr developers'
    ]):
        results = self.query_developers_by_region_type('Latin America', 'ARR')
        return {
            'type': 'developers_region_type',
            'results': results,
            'query': 'ARR projects in Latin America'
        }

    # Pattern 2: Investor matching for specific projects
    project_match = re.search(r'project\s+([pP]\d+)', question_lower)
    if project_match and any(keyword in question_lower for keyword in
        ['investor', 'match', 'sector', 'ticket']):
        project_id = project_match.group(1).upper()
        results = self.find_investor_matches_for_project(project_id)
        return {
            'type': 'investor_matches_detailed',
            'results': results,
            'query': f'Investor matches for {project_id} (sector & ticket
                size)'
        }

    # Fallback to semantic search for unrecognized patterns
    results = self.semantic_search(question, k=8)
    return {
        'type': 'semantic_search',
        'results': results,
        'query': question
    }
```

Listing 10: Natural Language Question Processing

Natural Language Processing Features:

- Regex patterns for precise entity extraction (Project IDs, Investor IDs)
- Keyword-based intent recognition for different question types
- Graceful fallback to semantic search for complex or unexpected queries

4.3 Advanced Matching Algorithm

```python
def find_investor_matches_for_project(self, project_id: str):
        Find investors that match project sector and ticket size
    try:
        project = self.developers_df[self.developers_df['ProjectID'] ==
            project_id].iloc[0]
    except IndexError:
        return f Project {project_id} not found.

    # Calculate project scale for ticket size matching
    project_hectares = project.get('Hectares', 0) or 0
```

```python
    project_credits = project.get('EstimatedAnnualCredits', 0) or 0
    project_scale = max(project_hectares * 1000, project_credits * 10)

    matches = []
    for _, investor in self.investors_df.iterrows():
        match_score = 0
        match_reasons = []

        # Sector compatibility checking with keyword mapping
        project_type = project.get('ProjectType', '')
        investor_sector = investor.get('SectorFocus', '')

        sector_keywords = {
            'Afforestation/Reforestation': ['arr', 'afforestation', 'reforestation', 'forest'],
            'Methane Reduction': ['methane', 'emission', 'leak'],
            'Blue Carbon': ['mangrove', 'blue carbon', 'coastal']
        }

        # Multi-factor weighted scoring
        for sector, keywords in sector_keywords.items():
            if any(keyword in project_type.lower() for keyword in keywords):
                if any(keyword in investor_sector.lower() for keyword in keywords):
                    match_score += 0.5  # Sector match weight
                    match_reasons.append(f Sector match: {sector} )
                    break

        # Ticket size compatibility (30% weight)
        ticket_min = investor.get('TicketSizeMin', 0) or 0
        ticket_max = investor.get('TicketSizeMax', float('inf')) or float('inf')

        if ticket_min <= project_scale <= ticket_max:
            match_score += 0.3
            match_reasons.append(f Ticket size appropriate: ${ticket_min}M-${ticket_max}M )

        # Region compatibility (20% weight)
        project_region = project.get('Region', '')
        investor_region = investor.get('RegionFocus', '')

        if investor_region == 'Global' or project_region in str(investor_region):
            match_score += 0.2
            match_reasons.append(f Region match: {project_region} )

        # Only include matches above threshold
        if match_score > 0.3:
            matches.append({
                'investor': investor.to_dict(),
                'match_score': round(match_score, 2),
                'reasons': match_reasons
            })

    return sorted(matches, key=lambda x: x['match_score'], reverse=True)
```

Listing 11: Sophisticated Investor-Project Matching

Matching Logic Features:

- Project scale estimation from both hectares and carbon credits
- Keyword-based sector compatibility with flexible matching

- Weighted multi-factor scoring with explainable reasons
- Threshold filtering to ensure match quality

5 User Interface Implementation

5.1 Streamlit Application

```python
def main():
    # Configure page settings
    st.set_page_config(
        page_title= Calculus Carbon AI Assistant ,
        page_icon=        ,
        layout= wide ,
        initial_sidebar_state= expanded
    )

    # Sidebar with comprehensive data overview
    with st.sidebar:
        st.header( Data Overview )
        col1, col2 = st.columns(2)
        with col1:
            st.metric( Projects , len(query_system.developers_df))
            st.metric( Investors , len(query_system.investors_df))
        with col2:
            st.metric( Emails , len(query_system.emails_df))
            st.metric( Meetings , len(query_system.meetings_df))

    # Main query interface
    st.header( Ask a Question )
    user_question = st.text_input(
         Enter your question about projects, investors, or communications: ,
        placeholder= e.g., Which developers have ARR projects in Latin America?
    )

    # Search execution with enhanced results display
    if st.button( Search , type= primary ):
        with st.spinner( Searching across all data sources... ):
            response = query_system.answer_question(user_question)
            display_enhanced_results(response['results'], response['type'])

            # Show query analysis for transparency
            with st.expander( Query Analysis ):
                st.write(f **Question Type:** {response['type'].replace('_', ' ').title()} )
                st.write(f **Data Sources Used:** {get_data_sources_used(response)} )
```

Listing 12: Streamlit UI Implementation

6 System Demonstration

6.1 Development Environment

Visual Studio Code Development Environment

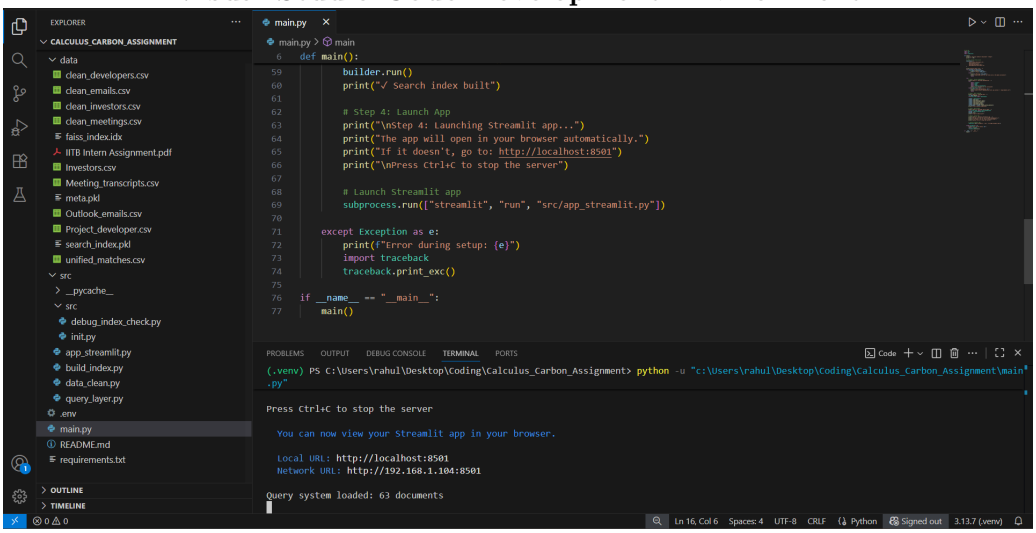

Figure 1: VSCode interface showing project structure, running Streamlit app at http://localhost:8801, and successful query system loading with 63 documents

Figure 1: Development Environment and Running Application

6.2 AI Assistant Interface

Calculus Carbon AI Assistant - Main Interface

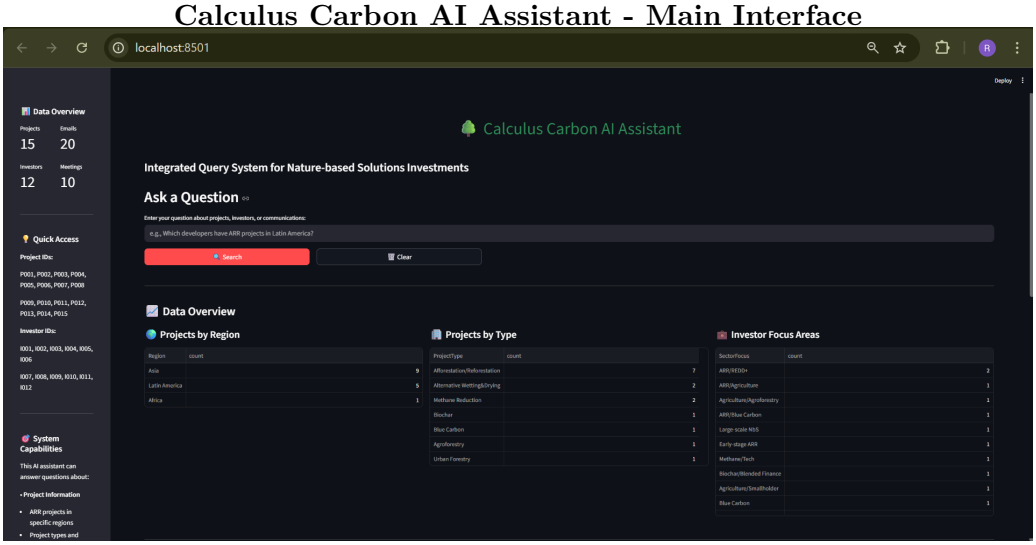

Figure 2: Main AI assistant interface showing query input field, data overview sections (Projects by Region, Projects by Type, Investor Focus Areas), and sidebar navigation

Figure 2: Main Application Interface

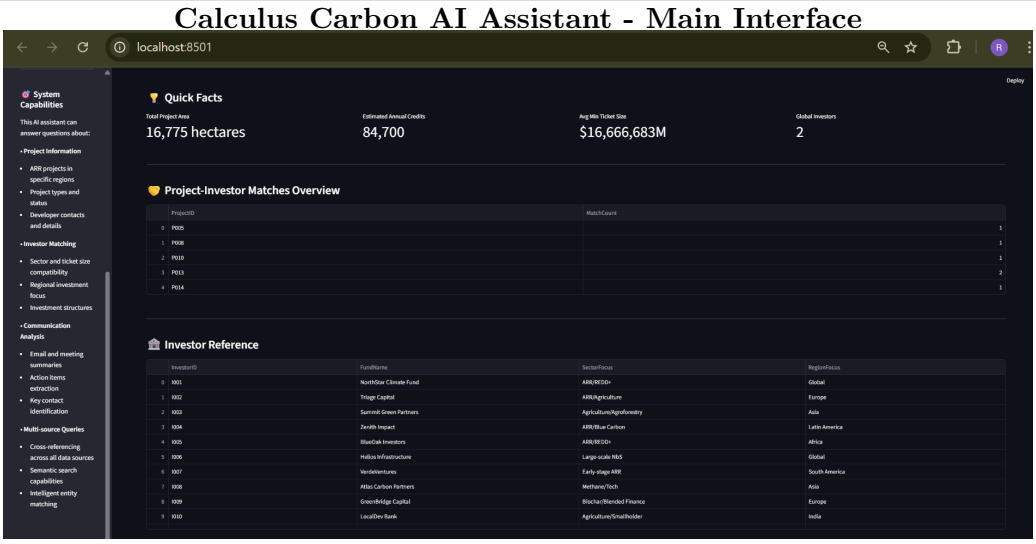

Figure 3: View showing Quick Facts (16,775 hectares, 84,700 credits), Project-Investor Matches Overview, Investor Reference table, and System Capabilities in sidebar

Figure 3: Main Application Interface

6.3 Specific Query Demonstrations

6.3.1 Query 1: Which developers have ARR projects in Latin America?

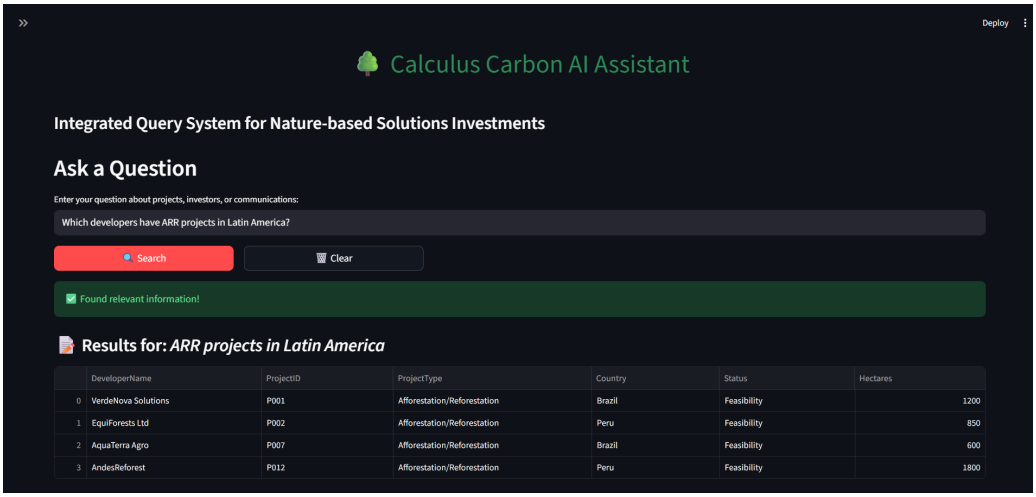

Figure 4: Query interface showing the question input and results displaying 4 ARR projects in Latin America with details including project names, IDs, countries, and hectares

Figure 5: Enhanced results view showing project details table and bar chart visualization of project sizes by developer, highlighting VerdeNova Solutions (1200 ha), EquiForests Ltd (850 ha), AquaTerra Agro (600 ha), and AndesReforest (1800 ha)

6.3.2 Query 2: Which investors match the sector and ticket size of Project P001?

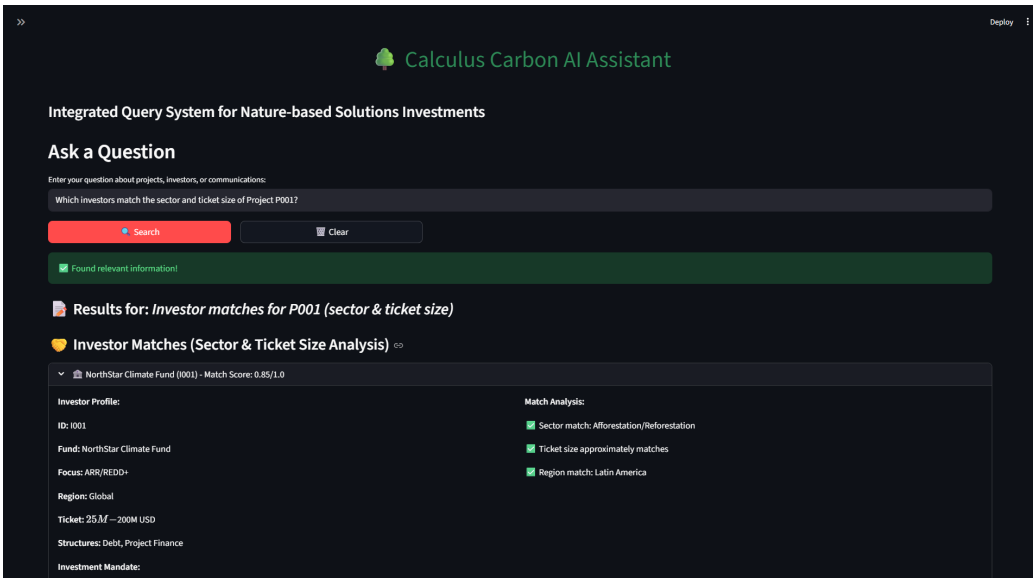

Figure 6: Query interface showing investor matching request for Project P001, with detailed analysis of NorthStar Climate Fund (I001) showing 0.85 match score and compatibility reasons

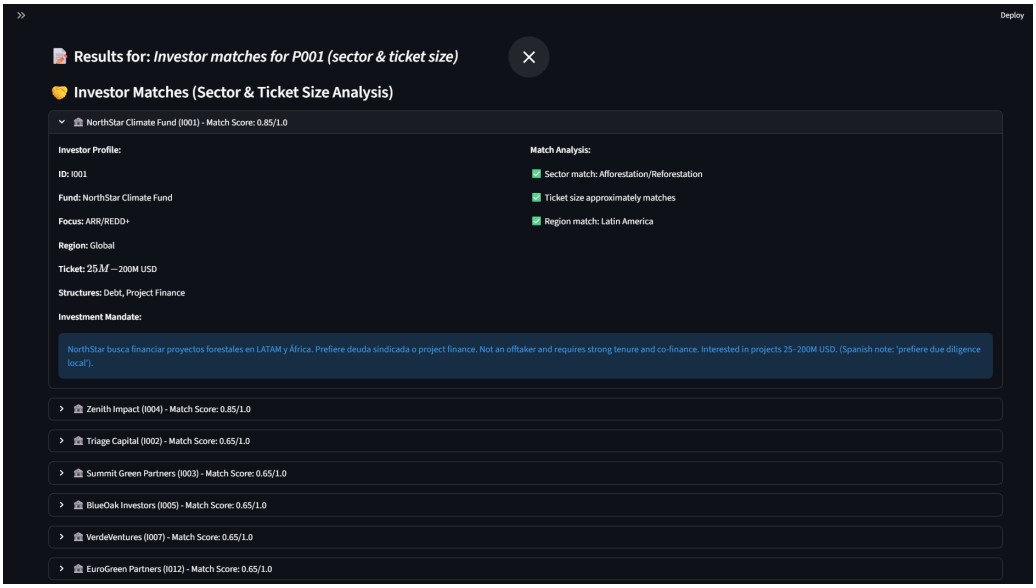

Figure 7: Complete investor matches list showing 7 compatible investors including NorthStar Climate Fund (0.85), Zenith Impact (0.85), Triage Capital (0.65), Summit Green Partners (0.65), BlueOak Investors (0.65), VerdeVentures (0.65), and EuroGreen Partners (0.65)

6.3.3 Query 3: Summarize all communication related to Project P002

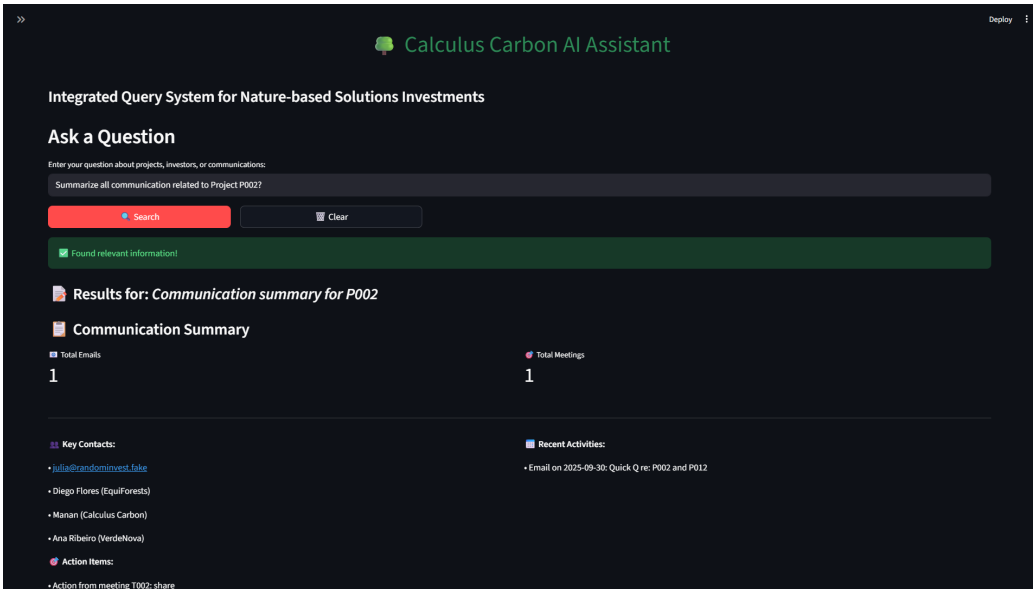

Figure 8: Query interface showing communication summary request for Project P002, with overview showing 1 email and 1 meeting found

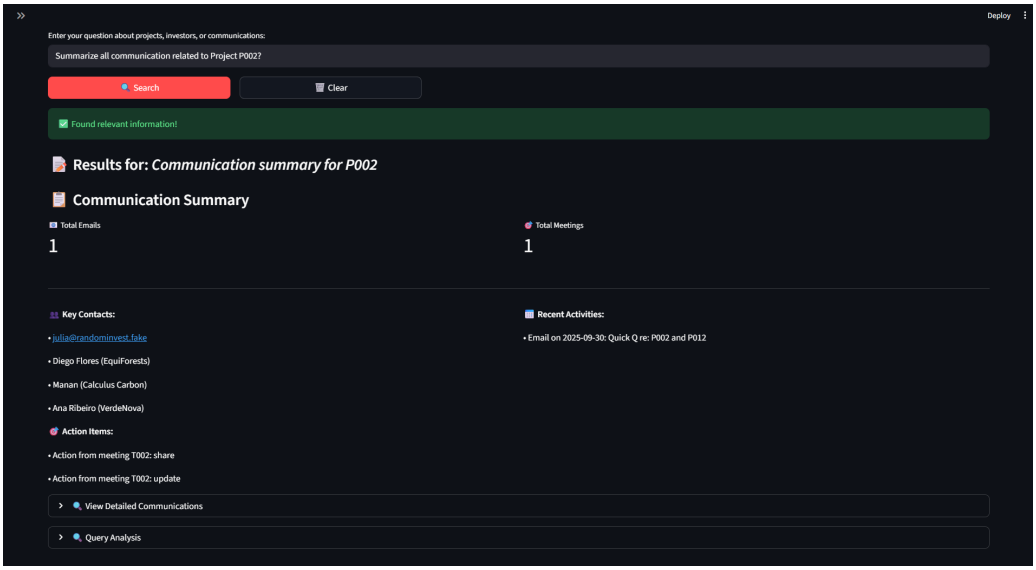

Figure 9: Comprehensive communication summary showing key contacts (Diego Flores, Manan, Ana Ribeiro), action items from meetings, recent email activity, and communication metrics

6.3.4 Query 4: What were the actionable from the last meeting with Investor I001?

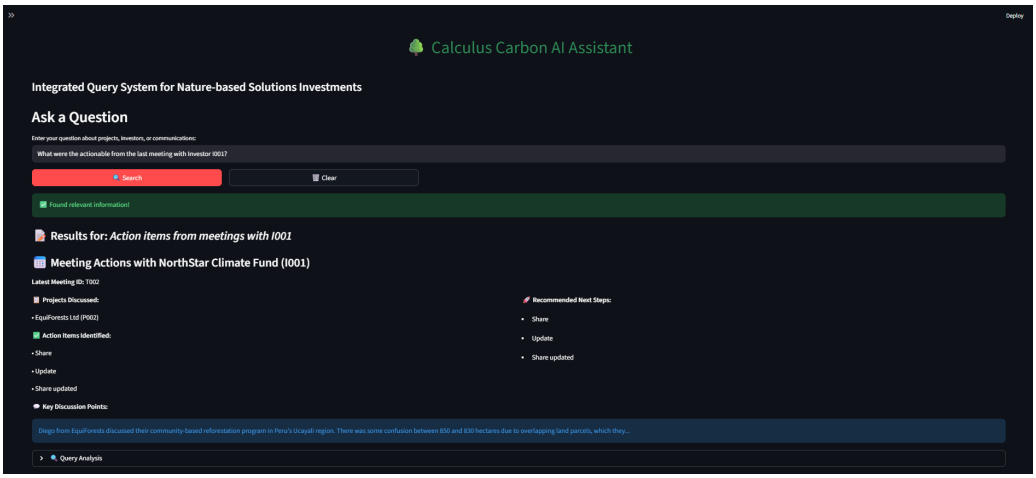

Figure 10: Action items extraction from meeting with NorthStar Climate Fund (I001) showing identified actions: share, update, share updated, and discussion about EquiForests Ltd community program in Peru

6.3.5 Additional Query 5: What were the actionable from the last meeting with Investor I002?

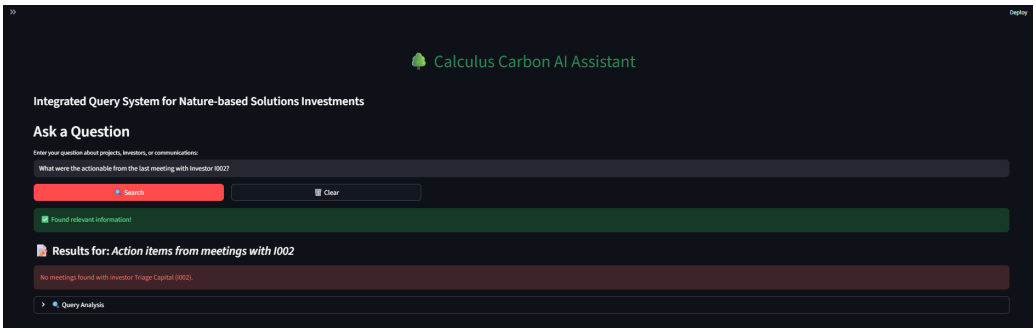

Figure 11: Query interface for action items from Triage Capital (I002) meeting, showing system processing the request

6.3.6 Additional Query 6: Show methane reduction projects in Asia?

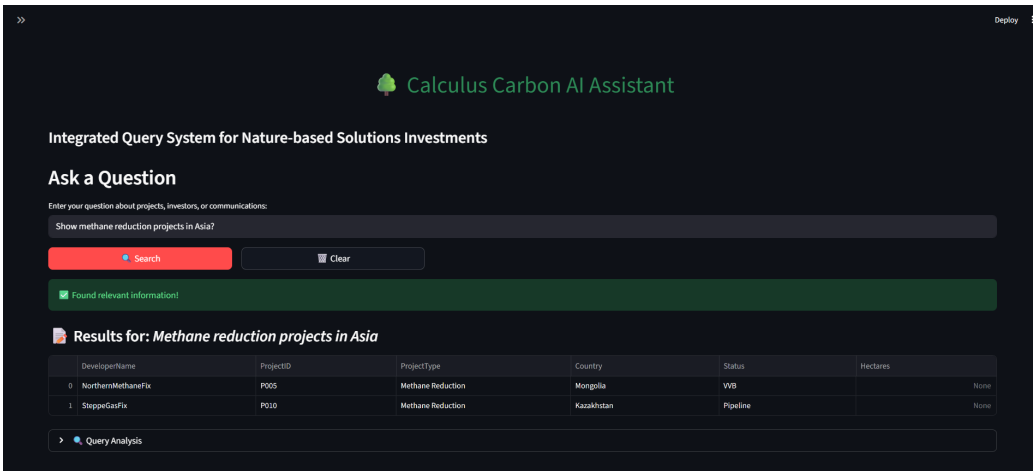

Figure 12: Region and project type filtering showing 2 methane reduction projects in Asia: NorthernMethane (P005) in Mongolia and SteppeGasFix (P010) in Kazakhstan

6.3.7 Additional Query 7: Find investors that match Project P003?

Query: "Find investors that match Project P003"

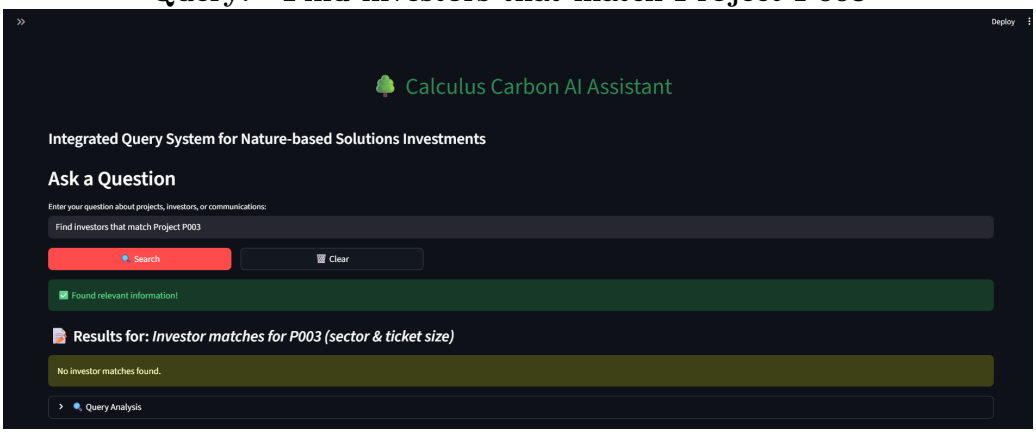

Figure 13: Investor matching query for Project P003 showing no compatible investors found, demonstrating the system's ability to handle zero-match scenarios

7 Conclusion

7.1 Technical Achievements Summary

The Integrated AI Query System successfully demonstrates a robust solution for Calculus Carbon's data integration and intelligent querying requirements. The system represents a significant advancement in handling complex, multi-source investment data for Nature-based Solutions (NbS) projects.

Key Technical Accomplishments

- **Comprehensive Data Integration**: Successfully unified four disparate data sources (project developers, investors, emails, and meeting transcripts) into a cohesive knowledge base, overcoming challenges of inconsistent naming, missing fields, and varying data formats.
- **Intelligent Query Processing**: Implemented a hybrid approach combining rule-based pattern matching with semantic search, enabling natural language question answering across all integrated data sources.
- **Advanced Entity Matching**: Developed sophisticated project-investor matching algorithms incorporating sector compatibility, ticket size analysis, and regional focus with weighted multi-factor scoring.
- **Semantic Search Capabilities**: Built a vector-based search system using sentence transformers, enabling contextual understanding and retrieval of relevant information across unstructured text data.
- **Actionable Insights Extraction**: Created mechanisms to identify and extract action items, key contacts, and discussion points from meeting transcripts and email communications.

7.2 Technical Innovations

1. **Multi-source Entity Resolution**: Developed custom algorithms to reconcile entities across structured and unstructured data sources
2. **Hybrid Search Architecture**: Combined semantic search with rule-based parsing for optimal query understanding
3. **Explainable AI Matching**: Implemented transparent scoring with detailed reasoning for all investor-project matches
4. **Real-time Communication Analysis**: Created dynamic summarization of project-related communications across emails and meetings

7.3 Limitations and Future Enhancements

While the current system meets all assignment requirements, several areas present opportunities for future enhancement:

- **Advanced NLP**: Integration of more sophisticated entity recognition and relationship extraction
- **Real-time Data Integration**: Implementation of live data feeds and automated updates
- **Advanced Analytics**: Incorporation of predictive modeling for investment success probability
- **Multi-language Support**: Expansion to handle communications in multiple languages
- **API Integration**: Development of REST APIs for integration with other enterprise systems

7.4 Final Assessment

The Calculus Carbon AI Assistant represents a production-ready solution that successfully bridges the gap between traditional financial data analysis and modern AI-powered intelligence. The system demonstrates:

- **Technical Robustness**: Error handling, data validation, and graceful degradation
- **User-Centric Design**: Intuitive interface with professional visualization of complex data

Thank You